Jia Wei TEH

CURRICULUM VITAE

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Research Interests

Stellar feedback, the dispersal of giant molecular clouds, and what interstellar bubbles reveal about the clusters and clouds that drive them. I designed and implemented TRINITY, an efficient 1D model of stellar winds, radiation and photoionised gas.

Education

PhD in Astronomy and Cosmology Jun 2022 - present
International Max Planck Research School for Astronomy and Cosmic Physics
University of Heidelberg, Heidelberg, Germany
Thesis: *Stellar feedback and the fate of giant molecular clouds*
Supervised by: Prof. Ralf Klessen, Prof. Simon Glover, Dr. Kathryn Kreckel
Expected PhD: by May 2027

BSc. (Adv.) (Hons.) in Astronomy & Astrophysics Feb 2018 - Nov 2021
Australian National University, Canberra, Australia

- Graduated with First Class Honours
- Thesis: *Constraining the escape fraction with LEGUS–SIGNALS survey*
- Supervised by Dr. Kathryn Grasha, Prof. Mark Krumholz, Dr. Andrew Battisti

First-author Publications

Teh, J.W., Klessen, R.S., Glover, S.C.O., Kreckel, K., 2026. *TRINITY: A coupled model of winds, radiation, and photoionised gas in molecular clouds. I. Methods and validation.* Astronomy & Astrophysics, 714, A226. doi:10.1051/0004-6361/202661074

Teh, J.W., Grasha, K., Krumholz, M.R., Battisti, A.J., Calzetti, D., Rousseau-Nepton, L., Rhea, C., Adamo, A., Kennicutt, R.C., Grebel, E.K., Cook, D.O., et al., 2023. *Constraining the LyC escape fraction from LEGUS star clusters with SIGNALS H* II *region observations: A pilot study of NGC 628.* Monthly Notices of the Royal Astronomical Society, 524(1), pp.1191-1210.

Co-authored Publications

Sheriff, K.E., Mills, E.A.C., Mai, X., Siwakoti, U., Duval, S.E., Bolatto, A.D., Levy, R.C., **Teh, J.W.**, et al., 2026. *JWST Observations of Starbursts: A Young Bubble in NGC 253's Central Starburst.* The Astronomical Journal, accepted. arXiv:2609.16190.
Contribution: TRINITY modelling of the bubble.

Soler, J.D., Beuther, H., Glover, S.C.O., Klessen, R.S., Ott, J., Rugel, M., **Teh, J.W.**, Clark, S.E., et al., 2026. *The Neutral Atomic Hydrogen in the solar neighborhood (NeAtHood) project: I. Ghost in the shell: Neutral atomic hydrogen in the extended Orion nebula.* Astronomy & Astrophysics, 711, A85.
Contribution: TRINITY model of the shell mass, compared with the H I and [C II] estimates.

Scheuermann, F., Kreckel, K., **Teh, J.W.**, Belfiore, F., Groves, B., Barnes, A.T., et al., 2026. *Stellar associations powering H* II *regions: II. Escape fraction of ionising photons.* Astronomy & Astrophysics, 708, A202.
Contribution: SLUG stellar population modelling.

Polak, B., Mac Low, M.-M., Klessen, R.S., **Teh, J.W.**, Cournoyer-Cloutier, C., Andersson, E.P., Appel, S.M., et al., 2024. *Massive star cluster formation: I. High star formation*

efficiency while resolving feedback of individual stars. Astronomy & Astrophysics, 690, A94.
Contribution: WARPFIELD models of the shell for a range of star formation efficiencies (Sect. 4.3).

Maconi, E., Soler, J.D., Reissl, S., Girichidis, P., Klessen, R.S., Hennebelle, P., Molinari, S., Testi, L., Smith, R.J., Sormani, M.C., **Teh, J.W.**, Traficante, A., 2023. *Modelling Local Bubble analogs: synthetic dust polarization maps.* Monthly Notices of the Royal Astronomical Society, 523(4), pp.5995-6010.

In Preparation

Teh, J.W. and the LVM collaboration. *Stellar feedback in the Rosette Nebula.*

Teh, J.W., Klessen, R.S., Glover, S.C.O., Kreckel, K. *What sets the fate of giant molecular clouds? II. A comprehensive study of interstellar bubbles in TRINITY simulations.*

Kabanovic, S., et al. (incl. **Teh, J.W.**). [O I] study of the M43 bubble with SOFIA/upGREAT.

Software

TRINITY (designed and written by me; lead developer): public 1D thin-shell stellar feedback code for interstellar bubble evolution. Documentation: jiaweiteh.github.io/trinity-web. TRINITY models appear in Soler et al. (2026) and Sheriff et al. (2026).

ASTROSIMPLIFY (designed and written by me): public Python package that downsamples 1D astrophysical profiles, such as radial profiles, stellar tracks and spectra, while keeping sharp transitions and extrema.

Submitted Proposals

- *Co-PI*, JWST Cycle 6 proposal 2026
- *Collaborator*, NASA ROSES-2025 Habitable Worlds Observatory Precursor Science Investigations proposal 2026

Talks

- Jan 2027: *Invited speaker*, "Orion & Beyond", Lorentz Center, Leiden, NL
- Aug 2026: Cologne–Bonn Symposium on the Habitats of Massive Stars across Cosmic Time, University of Cologne, DE
- 2026: SDSS-V Collaboration Meeting, KU Leuven, BE
- Jun 2023: 2nd IMPRS Seminar, Max Planck Institute for Astrophysics, Garching, DE
- Nov 2022: 1st IMPRS Seminar, Heidelberg University, DE
- Oct 2022: Local Volume Mapper Early Science Workshop, DE
- Dec 2021: Australian Institute of Physics Summer Meeting, Queensland University of Technology, AU
- Sep 2021: Australasian Council of Undergraduate Research Conference, Australian National University, AU

Posters

- Jun 2026: European Astronomical Society Annual Meeting, Lausanne, CH
- Jul 2025: 18th Potsdam Thinkshop, Leibniz Institute for Astrophysics Potsdam, DE
- May 2025: Celebration of John Bally: Star Formation, Stellar Feedback and the Ecology of Galaxies, Visegrád, HU
- Oct 2023: Harvard–Heidelberg Star Formation Workshop, Center for Astrophysics, Cambridge MA, US
- Nov 2021: 7th Australia–China Workshop on Astrophysics (ACAMAR7), AU

Awards, Honours and Grants

SDSS-V Early Career Researcher Travel Grant 2026
CSIRO and Swinburne summer research studentships 2021, 2020
Ida C. Koran Student Aid Scholarship Program (Undergraduate) 2018

Teaching and Communication

Teaching Assistant, University of Heidelberg, Germany

- Introduction to Computational Physics 2023
- MVComp: Fundamentals of Simulation Methods 2022

Science communication and outreach

- *Invited talk*, "How to Effectively Use Colours", early-career session, EAS 2026
- *Invited talk*, "Colour Theory and Visualisation", HGSFP Winter School 2025
- *Lecturer*, Girls' Day, Physics and Astronomy, Heidelberg University 2024, 2025
- *Talk*, "Matplotlib: Tips, tricks, and common pitfalls" 2024
  IMPRS Astro Hackathon, Haus der Astronomie, Heidelberg
- *Guest author*, Astrobites Undergraduate Research series Oct 2021
  The mystery of orphan H II *regions: where are the ionising star clusters?*

Organising Committees

- *LOC*, GMAC2: The Global Malaysian Astronomers Convention 2027
- *SOC*, Early-career session, EAS Annual Meeting 2024, 2025
- *Co-chair*, Institute for Theoretical Astrophysics Journal Club 2023
- *LOC*, International Astronomical Union Symposium (IAUS) 377 2022

Collaborations and Memberships

- PHANGS collaboration
- SDSS-V Local Volume Mapper
- ECOGAL ERC Synergy consortium
- European Astronomical Society (EAS)
- Astronomische Gesellschaft (German Astronomical Society)

Undergraduate Research

- 2021–22: Galaxy-cluster masses with deep learning; CSIRO (N. Gupta)
- 2021: HCN emission in the Antennae; LEAPS, Leiden Observatory (A. Bemis)
- 2020–21: Black-hole binaries and strong emission lines; Swinburne (T. Nanayakkara)
- 2020: Star clusters and H II regions in NGC 628; ANU (K. Grasha, A. Battisti)

Observing Experience

Jan 2022: *Duty astronomer*, Australia Telescope Compact Array, radio candidates from the FLASH survey (6 h)

References

- Ralf Klessen klessen@uni-heidelberg.de
  Full Professor, University of Heidelberg
- Kathryn Kreckel kathryn.kreckel@uni-heidelberg.de
  Research Group Leader, University of Heidelberg
- Mark Krumholz mark.krumholz@anu.edu.au
  Full Professor, Australian National University